Restoration Ecology
THE JOURNAL OF THE SOCIETY FOR ECOLOGICAL RESTORATION

RESEARCH ARTICLE

Top-down control of rare species abundances by native ungulates in a grassland restoration

Brian J. Wilsey[1,2], Leanne M. Martin[1,3]

Ecological restorations are predicted to increase in species diversity over time until they reach reference levels. However, chronosequence studies in grasslands often show that diversity peaks after the first few years and then declines over time as grasses become more dominant. We addressed whether bison grazing and seed additions could prevent this decline in diversity. Exclosures that prevented bison grazing were compared with grazed plots over 4 years, and seed additions were conducted inside and outside exclosures to test for seed and microsite limitations. A previous study conducted 4-months post seeding found that local species richness was primarily seed limited, but that grazing could sometimes increase seedling emergence. Here, we tested whether increased seedling emergence led to longer-term increases in the species diversity of the plant community. We found that the seed addition effect grew smaller and the grazing effect grew stronger over time, and that seed additions affected the abundance of added species only when plots were grazed. Grazed plots had higher species diversity and lower biomass and litter buildup compared to non-grazed plots. Our results suggest that moderate grazing by bison or management that mimics grazing can maintain diversity in grass-dominated situations. Our results also emphasize the need to follow seed additions over several years to assess correctly whether seed limitation exists.

Key words: ***Bos bison*, Iowa, seed limitation, species diversity, tallgrass prairie, ungulate grazing**

Implications for Practice

- Introduction of bison into grassland restorations can lead to rare plant species recruitment when seeds are present.
- Mimicking the effects of bison, with concurrent seed additions might be an effective way of increasing the diversity of grass-dominated restorations.
- Seed addition studies should follow plant recruitment over several years before determining whether a site is seed limited.

Introduction

Restoring appropriate levels of species diversity is a common objective of restorations, and diversity is often used as a measure of success (Martin et al. 2005; Polley et al. 2005). It is commonly assumed that species diversity in grassland will rise to reference levels given enough time following seeding of native species in former croplands (Fig. 1, green line). However, monitoring often finds that diversity and richness decline after reaching a peak in the first 3–5 years due to increasing perennial grass dominance (e.g. Sluis 2002; Camill et al. 2004; Grman et al. 2013; but see Kucharik et al. 2006, Fig. 1 red line). Testing whether diversity will continue to rise until reaching the levels of reference sites, or if it will reach a maximum value at intermediate stage of development (and then drop) remains one of the most important questions for restoration ecologists to answer. Developing a better understanding of how to prevent the drop in diversity will enable restorationists to improve existing low diversity restorations.

Low species diversity in heavily grass-dominated restorations is hypothesized to be due to (1) seed limitation, (2) microsite limitation that prevents recruitment of seedlings, or (3) a combination of both (Eriksson & Ehrlen 1992; Tilman 1997; Zobel et al. 2000; Foster 2001; Foster et al. 2004; Martin & Wilsey 2006). Restorations are often conducted in fragmented patches that are not linked to species-rich seed sources (Damschen et al. 2006). Seed additions can increase rare species abundances and species diversity in some cases (Smith et al. 2000; Pywell et al. 2002), especially if the site is one of relatively low productivity (Foster 2001). If seed limitation is the dominant constraint, then adding seeds of rare or missing species should increase diversity even in systems with few available microsites. Alternatively, if microsites are limiting, then adding seeds will have little effect (Howe 2000; Sluis 2002; Camill et al. 2004; Dickson et al. 2009). Dominance by C_4 grasses in grassland restorations, which occurs as soon as 3 years after initiating tallgrass prairie restorations (Camill et al. 2004), can become very high over time (Baer et al. 2002, 2004). Grass dominated areas can have large canopies and thick litter layers that reduce light and water (Xiong & Nilsson 1999), or they can have lawn-like conditions

Author contributions: BW conceived the project; BW, LM designed the research and performed the experiments; BW analyzed the data; BW, LM wrote the paper.

[1]Department of Ecology, Evolution and Organismal Biology, Iowa State University, Ames, IA 50011, U.S.A.
[2]Address correspondence to B. J. Wilsey, email bwilsey@iastate.edu
[3]Present address: Kansas Biological Survey, University of Kansas, Lawrence, KS, U.S.A.

doi: 10.1111/rec.12197

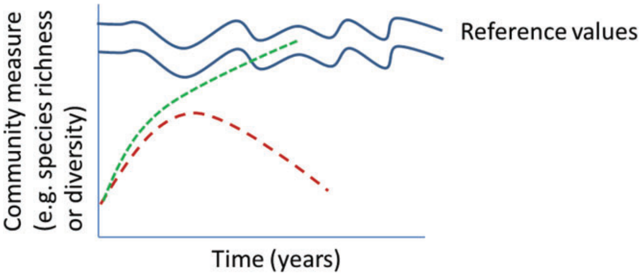

Figure 1. Theoretical figure describing how community measures (e.g. species diversity) are predicted to change over time during a restoration. The first line (green dashed) denotes a continuous increase in values until values fall within the range of remnant areas. The second line (red dashed) suggests that values will peak with an intermediate amount of time, and then values decline as dominance by grasses increases (supported by chronosequence study of Sluis (2002)). Moderate grazing in this study was predicted to prevent the drop in species diversity, and grazing and seed additions were predicted to lead to an increased trajectory.

due to high and dense rhizome growth (Wilsey 2010), both of which can lead to low seedling survival in establishing species (Henry et al. 2004). Management practices such as frequent spring burning and removal of grazing mammals can sometimes exacerbate C_4 grass dominance (Collins et al. 1998; but see Bowles & Jones 2013). Thus, if the microsite limitation hypothesis is the primary reason for low diversity, then reducing grass dominance should lessen competitive suppression of rare species, increase seedling establishment, and eventually, increase plant diversity (Foster & Gross 1997; Knapp et al. 2012). This may act in concert with seed limitations, that is, both hypotheses may be supported (Turnbull et al. 2000; Foster & Dickson 2004; Martin & Wilsey 2006).

Disturbances such as fire and grazing can be important regulators of species diversity in many grassland systems (Collins et al. 1998; Allred et al. 2011*b*; Collins & Calabrese 2012; Koerner & Collins 2014). Thus far, restoration ecologists have stressed the importance of fire much more than native ungulate grazing. However, understanding the role of grazers is becoming necessary as bison (*Bos bison* L.) or elk (*Cervus elaphus* L.) are reintroduced (Knapp et al. 1999; Martin & Wilsey 2006; Jackson et al. 2010; Kohl et al. 2013), and cattle are used to facilitate establishment of native grasses and forbs in restorations (Bouressa et al. 2010; Allred et al. 2011*a*). Although very intense grazing can be detrimental to species diversity, moderate grazing (either in frequency or intensity) could have positive effects on species diversity when the grazing mammals feed on dominant grasses and sedges, especially when grazing is patchy (Allred et al. 2011*b*). Moderate grazing can increase plant species diversity by increasing light or water availability in grazed stands (McNaughton 1979; Hartnett et al. 1996; Collins et al. 1998; Knapp et al. 1999). In a restoration context, adding native grazing mammals could lead to greater seedling emergence (Martin & Wilsey 2006) and recruitment of rare plant species, and this could reverse the trend of declining species diversity over time. We tested this idea in this study.

In this manuscript, we build off a previous study (Martin & Wilsey 2006) that looked at seedling emergence following seed additions to plots in a large restoration project in Iowa, U.S.A. Here we present data on the longer-term effects of grazing as well as how these seedlings eventually affected species diversity and rare species establishment. Seedling emergence without successful recruitment will not affect species diversity or any other community-level measure in the long term (Wilsey & Polley 2003), which necessitates following seedlings to determine if they successfully produce perennating organs such as rhizomes and crowns that can survive the winter, or if they can successfully reproduce sexually. Only then can we say for sure that diversity will be affected in the long term. The objectives of our experiment were to determine: (1) whether seed additions, native ungulate grazing, or both can enhance species diversity in highly grass-dominated tallgrass prairie restorations, and (2) if native ungulates increase "rare" native prairie species. Previously, we found that these plantings had lower small-scale species richness than remnants (Martin et al. 2005). We report here on the restored plant communities during years 9–12 to determine if species diversity is increasing or decreasing over time, and if grazing is altering this trajectory.

Methods

Study Site

The study was done in a large prairie restoration project at the Neal Smith National Wildlife Refuge (NS) in Jasper County, IA, U.S.A. (41°33′N, 93°17′W). The objective of the project, which was begun in 1991 by the U.S. Fish and Wildlife Service, was to restore a large tallgrass prairie ecosystem on former crop fields using locally collected seeds. At the time of our study, the refuge spanned 2,104 ha, approximately 1,200 hectares of which was seeded with tallgrass prairie species, beginning in 1992. *Bos bison* and *Cervus elaphus* were introduced to a 303-ha enclosure in 1996 and 1998, respectively, which is where our study took place (Fig. 2). Approximately 30–50 *B. bison* and 15 *C. elaphus* inhabited the area during the years of our study. Grazing of our plots was primarily from bison and not by elk, based on the evidence of only bison dung in and near our grazed plots (personal observations), and the findings that elk spent most of their time near wooded creeks (Kagima & Fairbanks 2013). There were 20 separate plantings in this area (mean of approximately 14 ha each); each planted with separate bulk seed mixes collected from local prairie remnants. Management after seeding included yearly spring burning during the first few years followed by 2-year fire rotations, which is common for restorations (Packard & Mutel 1997; Copeland et al. 2002), and mowing when necessary to control weedy and invasive species (Pauline Drobney, personal communication). Our plots were neither burned nor mowed in 2003 or 2004, and all plots were burned in spring 2005. Further details are included in Martin and Wilsey (2006).

We randomly selected eight plantings (blocks) within the enclosure that were seeded between 1994 and 1996, using only plantings on formerly cropped areas to standardize sampling (Fig. 2). Thus, by the end of this study (2006), the plantings were

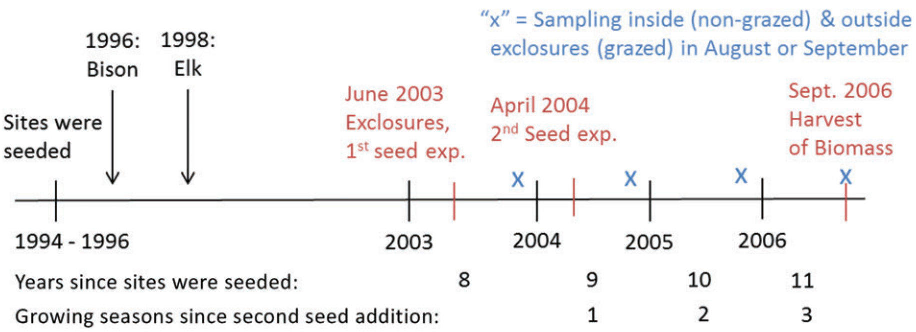

Figure 2. Timeline for restorations and sampling regimes. Seed additions experiments (exp.) were conducted by adding 10 and 25 species to separate subplots within exclosures and grazed plots. Seedlings were counted and analyzed monthly during 2004 in an earlier study (Martin & Wilsey 2006). Restorations were burned on a 2-year cycle before the study started in 2003. During the 2003–2006 time period, all plots were burned in April 2005.

10–12 years old. Four of the plantings included 6.7 kg/ha of *Elodea canadensis* as a cover crop and four did not (see Martin et al. 2005 for more on the effects of this cover crop).

Experimental Design

A randomized block design was employed, using the eight plantings (sites) as blocks. In June 2003, we established a 6 × 8 m exclosure within each site. Two 6 × 8 m grazed plots were established on either side of each exclosure in June 2003 with 5-m alley-ways between them. Twice as many grazed plots were used because of the more variable effects of grazing. By request of refuge staff, exclosures were kept out of view of visitors when possible, and this precluded completely random locations within randomly selected plantings. The 8 exclosed and 16 grazed plots were sampled over time as explained below to assess long-term effects of grazing.

The seed addition study was done with a split-plot design, with the grazing treatments applied to the main plots and seed addition treatments applied to subplots (i.e. to plots within each exclosure). Two seed addition treatments were applied to separate randomly located 1 m^2 subplots within each main plot (i.e. within each grazed or non-grazed plot) and compared to a control plot (not seeded) to assess the effects of grazing and seed additions. Seeds of 10 locally rare native forb and grass species from local remnants were added in the first addition in June 2003, and this was increased to 25 species in a second seed addition in April 2004 using seeds purchased from a local seed company (Allendan Seed Co., Winterset, IA, U.S.A.). The first seed addition led to very few emerging seedlings during the first 2 years (Martin & Wilsey 2006) and we found no additional seedlings in monitoring during this study (data not shown). Thus, we stress the second seed addition in this paper. Species added were *Bouteloua curtipendula*, *Sporobolus asper*, *Solidago speciosa*, *Pycnanthemum virginianum*, *Dalea purpurea*, *Chamaecrista fasciculata*, *Amorpha canescens*, *Lespedeza capitata*, *Monarda fistulosa*, *Eryngium yuccifolium* (both experiments), *Potentilla arguta*, *Silphium laciniatum*, *Echinacea pallida*, *Ratibida pinnata*, *Artemesia ludoviciana*, *Liatris pycnostachya*, *Verbena stricta*, *Helianthus rigidus*, *Gentiana andrewsii*, *Tradescantia bracteata*, *Viola pedatifida*, *Anemone cylindrica*, *Phlox pilosa*, *Schizachyrium scoparium*, and *Solidago rigida* (second experiment only, nomenclature follows Eilers & Roosa 1994). Seeds were added with equal number of seeds per species at a very high rate of 19,700 seeds/m^2, based on typical seed rain rate for tallgrass prairie (Rabinowitz & Rapp 1980). Thus, with the 72 subplots, we were testing whether realistically high seed numbers would increase seedling emergence and eventually, species diversity.

We estimated grazing intensity by comparing biomass inside and outside of temporary moveable exclosures (3 × 4 m) during each year of the study (McNaughton et al. 1996; Wilsey et al. 2002). Grazing intensity was reported in Martin and Wilsey (2006) as 14% of net primary productivity in spring and 49 and 68% of net primary productivity over two summers.

Long-term Sampling of Exclosures and Grazed Plots

Sampling over time within the grazed and exclosed plots was done over four growing seasons to determine the long-term effects of grazing on plant communities (Fig. 2). We estimated aboveground biomass and litter to compare vegetation structure between grazed and exclosed plots. Biomass and litter were measured because of their potential effects on seedling emergence (e.g. Grime 1973; Howe 2000; Fraser et al. 2014). Aboveground biomass was clipped to 2 cm in a 40 × 100 cm quadrat from random locations inside each exclosure or grazed plot, and litter on the soil surface was collected on each sampling date. Biomass was sorted into live (green present) and standing dead components; live material was then sorted by species, dried for 48 h at 65°C, and weighed. Species diversity calculations were done on biomass by species data at the quadrat scale for each grazed and exclosed plot. Diversity was quantified with Simpson's diversity (1/*D*), where $D = 1/\sum p_i^2$, and p_i = relative biomass of each species i, and with species richness (*S*, number of species per quadrat). Plots were sampled at peak biomass (Fig. 2).

Final Harvest of Seed Addition Experiment

In order to test whether emerging seedlings affected plant diversity and rare species recruitment after multiple growing

Table 1. ANOVA results ($F[p]$) for species richness and diversity, aboveground biomass, and litter, and combined biomass + litter in plots grazed by bison and non-grazed plots (inside 6 × 8 m exclosures) in a restoration at Neal Smith National Wildlife refuge, Iowa, U.S.A. Seed additions were conducted in subplots within each exclosed or grazed plot (control, seed addition 1, seed addition 2), and contrasts were between unseeded and seeded plots in seed addition 2 (see text for more details). Significant effects are in bold. Time zero values were included as a covariate for richness and Simpson's diversity.

Term	df	*Richness*	*Simpson's (1/D)*	*Litter (g)*	*Biomass (g)*	*Litter + biomass*
Long-term responses in main plots						
Graze (G)	1, 15	**11.2 (0.004)**	**4.8 (0.045)**	**6.8 (0.019)**	**4.0 (0.065)**	**7.7 (0.014)**
Time (T)	3, 65	**3.8 (0.014)**	**3.4 (0.024)**	**4.3 (0.008)**	6.8 **(0.001)**	0.1 (0.980)
G × T	3, 65	2.5 (0.067)	1.1 (0.34)	3.2 **(0.029)**	**3.9 (0.013)**	0.4 (0.738)
Time 0	1, 65	**7.1 (0.001)**	**15.4 (0.01)**			
Seed addition experiment						
Graze (G)	1, 13	**5.0 (0.04)**	**4.4 (0.05)**			
Seed (S)	2, 38	2.2 (0.13)	0.1 (0.89)			
G × S	2, 38	1.6 (0.21)	1.0 (0.39)			
Contrasts in seed addition 2						
S vs. control	1, 38	**4.3 (0.04)**	0.2 (0.63)			

df, degrees of freedom.

seasons, we clipped and sorted aboveground biomass by species at peak biomass in September 2006 (Fig. 2). Plots were clipped to the soil surface in a 40 × 100 cm quadrat placed in the center of each subplot. Thus, we were sampling 30 months after seeds were added in the second experiment. Species richness (S) and Simpson's diversity ($1/D$) were calculated in each subplot to determine if seed additions or grazing affected species diversity. Because the species that we used in our seed additions also could have been found in the control plots, we evaluated recruitment by comparing the biomass of seeded species in seeded and control plots. To perform this, an "enhancement effect" was calculated as ln[(species biomass in addition plots + 1)/(species biomass in control plots + 1)]. This estimates the magnitude of the increase in abundance of target species with seed additions and is in ln units (thus, to interpret the effect size, one must take the antilog).

Statistical Analyses

For the long-term comparisons of grazed and exclosed plots a repeated measures mixed model analysis of variance (ANOVA) (PROC MIXED of SAS; Littell et al. 2012) was used with grazing as the fixed effect and planting as the random effect. The repeated statement used an AR(1) covariance structure. We compared the first and last date with an a priori contrast (CONTRAST statement in SAS software) to test for changes in richness and diversity over time. Variables were *square-root* transformed (biomass and litter), which successfully normalized the data and reduced heteroscedasticity.

For the final harvest data from the seed addition experiment, we used split-plot ANOVA and analysis of covariance (ANCOVA). Grazing effects were tested with main plot error term (planting × grazed), and seed and seed × grazed interaction were tested with subplot error term. Comparisons of seeded and non-seeded plots were made with a priori contrasts. We used time 0 data (measurements taken before exclosures were constructed) as a covariate in the diversity and richness analyses, and then dropped it from each model if it was not significant ($p > 0.15$). To test whether seed additions had an enhancement effect on rare plant species, we compared the enhancement effect to 0 with a t-test. This was based on the null hypothesis that if seed additions had no effect on rare species, this value would be 0.

Results

Effects of Grazing Through Time

Mean species richness at the quadrat scale was higher, on average, when plots were grazed than when they were not (Table 1; Fig. 3A). Richness was higher in grazed plots consistently across dates (i.e. without a significant interaction with time). Richness did not increase over time during the time course of the experiment (contrast of first and last dates, $F_{[1,64]} = 0.13$, $p = 0.72$). In grazed areas, richness remained fairly constant over time, varying from a mean of 10.4–12.6 in the first two August dates to 10.4–11.4 in the last two dates (Fig. 3A). Richness was significantly lower in non-grazed (exclosed) plots, ranging from 9.6–9.9 in the first two August dates to 7.0–9.1 in the last two dates (Fig. 3). Simpson's diversity followed similar trends with grazing and time, with significantly higher diversity in grazed plots (Fig. 3B). Simpson's diversity marginally increased over time (contrast of first and last dates, $F_{[1,65]} = 3.9$, $p = 0.052$).

Litter and aboveground biomass were often higher inside exclosures than outside (Fig. 4), suggesting that seedling suppression and aboveground competition might have been responsible for species diversity differences. Extremely high aboveground biomass on the 27-month sampling date led to a large build-up of litter and a collapse of biomass by the 39-month sampling date (Fig. 4A & B). Litter was higher inside exclosures on all sampling dates except for the year that all plots were burned in the spring. That year had the greatest difference in aboveground biomass (Fig. 4B). Biomass + litter tended to average out these temporal fluctuations caused by fire-induced consumption of litter, and it was consistently higher inside exclosures than in grazed plots, leading to a significant main effect of grazing with no interactions (Table 1; Fig. 4C).

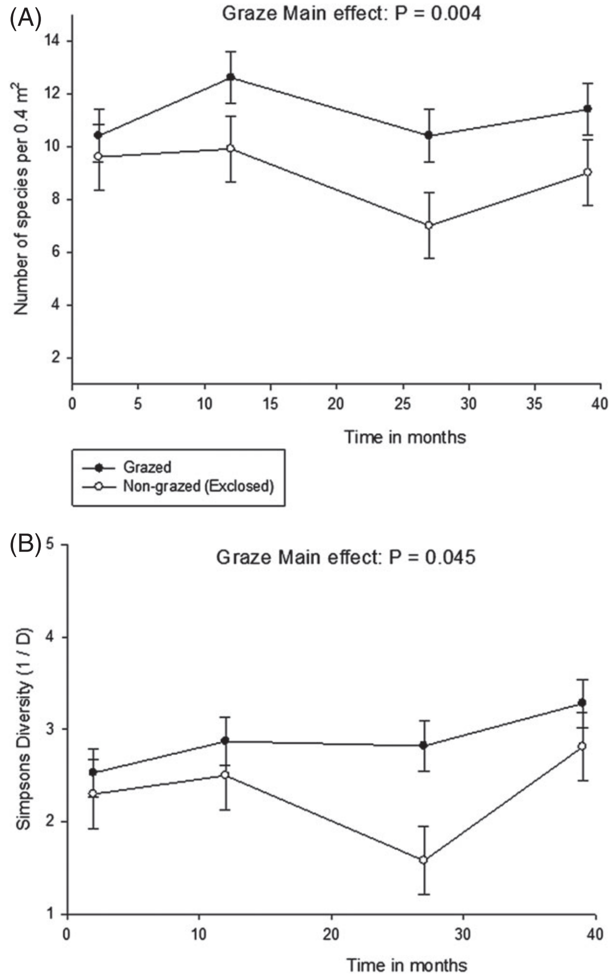

Figure 3. Species diversity measures (richness [A] and Simpson's diversity [B]) in grazed areas and non-grazed (exclosed) areas during a developing restoration at Neal Smith National Wildlife Refuge in Iowa (least-squares means ± SE from repeated measures mixed model ANOVA). "Fire" denotes the time period that a prescribed fire treatment was applied to all plots in the study.

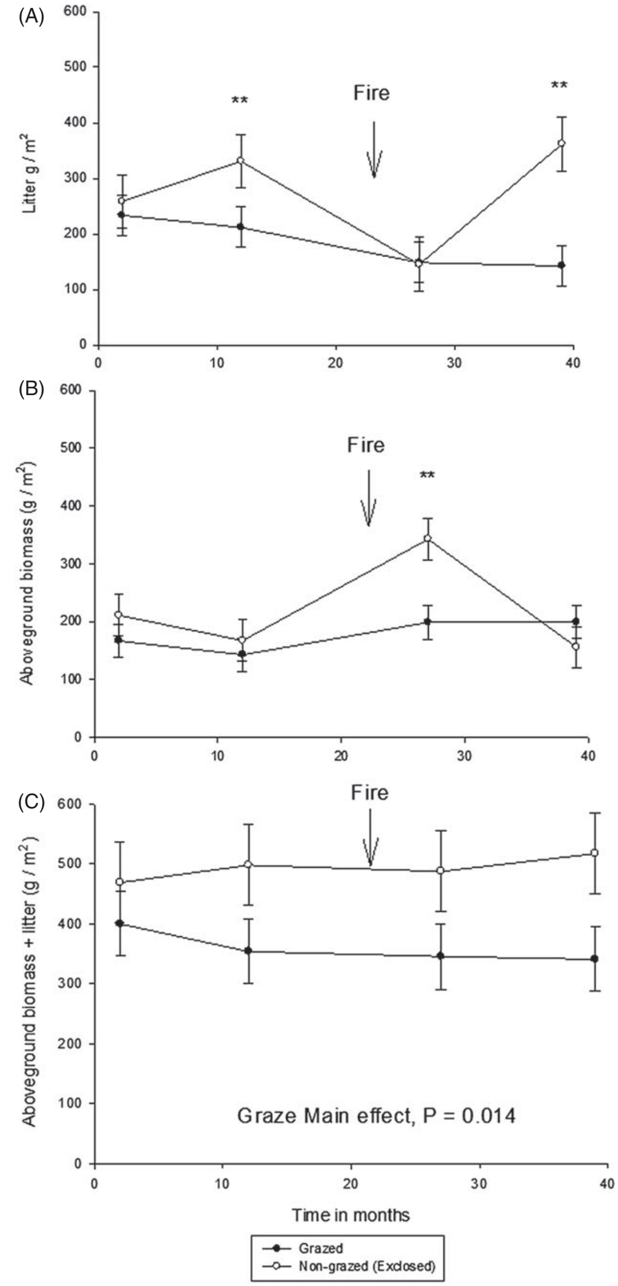

Figure 4. Litter (A), aboveground biomass (B) and litter + biomass (C) in grazed areas and non-grazed (exclosed) areas during a developing restoration at Neal Smith National Wildlife Refuge in Iowa (least-squares means ± SE from Repeated measures mixed model ANOVA). "Fire" denotes the time period that a prescribed fire treatment was applied to all plots in the study, and ** denotes $p < 0.05$ for a given time period.

Seed Addition Experiment — Final Harvest

Species Diversity. Grazing had larger and more consistent effects on species diversity and richness at the final harvest date than did seed additions. Grazed plots had higher species richness (11.4 + 0.87 vs. 9.4 + 0.99) and Simpson's diversity (2.89 ± 0.27 vs. 2.09 ± 0.35) than did non-grazed plots (Table 1). Seed additions did not significantly affect species richness (seed treatment: $F = 2.2$, $p = 0.1284$), with the contrast between the control and second seed addition being significant ($p = 0.044$). Species richness averaged 9.8 (standard error [SE] = 0.91) in control plots, and 10.7 (SE = 0.91) in seed addition treated plots. Seed additions also did not significantly alter Simpson's diversity and contrasts between controls and seed addition plots were non-significant (Table 1).

Rare Native-species Establishment. Rare species enhancement was found only in the second seed experiment, and only when plots were grazed (Fig. 5). Adding seeds led to a 4-fold ($e^{1.395} = 4.03$) increase in targeted species establishment compared with control plots in grazed plots ($t = 2.48$, degrees of freedom [df] = 7, $p = 0.042$). Non-grazed plots had a non-significant enhancement value ($t = 0.35$, df = 7, $p = 0.735$; Fig. 5). Species that were enhanced were *Chamaecrista fasciculata*, *Pycnanthemum virginianum*, *Ratibida pinnata,* *Silphium laciniatum*, *Echinacea pallida, Lespedeza capitata*, and *Monarda fistulosa.*

Discussion

We found that grazing had important effects on plant restorations that were in their later stages of development. Grazing maintained species diversity at a higher level than non-grazed plots during the 9–12 year time-frame of plantings. This occurred because there was less build-up of aboveground biomass and litter in grazed plots. Grazing effects on litter and biomass are well understood from bison grazing in intact systems, but they are less well understood for restorations. We found that grazing facilitated the establishment of rare species that were added in a seed addition. Importantly, we found that seed additions without grazing had no significant effect on rare species recruitment. This contrasts with our results from 4 months after seeding (Martin & Wilsey 2006), where we found that seed limitation was more important than grazing in rare species establishment. Following the fate of seedlings over the following years led to a much stronger effect of grazing and a reduction in the simple seed addition effect. This suggests that we need to follow seed addition experiments for sufficient time to completely answer questions about seed limitation (Wilsey & Polley 2003).

Rare species must successfully reproduce if they are to have any long-term effect on community composition or species diversity (Wilsey & Polley 2003). We found that recruitment results differed in important ways from seedling emergence results. Most importantly, the seed addition main effect alone weakened and became non-significant over time, and the importance of grazing strengthened (i.e. compare current results with those of Martin & Wilsey 2006). This suggests that it is recruitment into reproductive populations that are the most sensitive to grazing, and that seed additions to restorations may not always have positive effects on recruitment in the long run without associated disturbance. Martin and Wilsey (2014) found that seed additions into well-established non-grazed stands did not alter diversity or the proportion of exotic species, which suggests that community states may be hard to change once species are well established. Our results are based on sampling at the local plant neighborhood scale (i.e. 0.4 m^2), and the differences might appear to the reader to be fairly small. However, it is important to remember that small differences at the neighborhood scale can scale-up to be substantial at larger scales such as the field scale.

Grazing by bison and other ungulates are fairly well studied in intact systems (e.g. Knapp et al. 1999; Towne et al. 2005; Bakker et al. 2006; Allred et al. 2011*b*; Knapp et al. 2012; Lezama et al. 2014). Bakker et al. (2006) and Lezama et al. (2014) found that large grazing mammals had a greater effect on plant diversity in productive grassland sites such as ours. However, the effects of bison on restorations are poorly understood. The issue in restorations that differs from intact systems is that species must establish from seed in restorations, and seedling recruitment can be problematic in grass-dominated stands. We found here that grazing can have important effects on grass-dominated prairie restorations. Grazing was found to prevent the lower species richness that occurred in non-grazed plots. This suggests that large ungulate grazing or management that mimics grazing may be useful to maintain species richness in restoration plantings. However, it is important to keep in mind that positive effects of bison on rare species recruitment were only found when seeds were added. Grazing did not lead to recruitment of any plant species without seed additions, and the first seed addition had no recruitment whatsoever. Further work on the role of top-down processes is necessary in restoration ecology. We studied native ungulates, but most grazed grasslands are grazed by cattle (*Bos primigenius*) or other non-native livestock. Cattle are similar but not identical to bison when managed similarly (Allred et al. 2011*a*, 2011*b*). Cattle also prefer to graze on dominant grasses, but they have a slightly greater proportion of forbs in their diet compared to bison (Towne et al. 2005). This suggests that they might have similar but perhaps slightly lesser effects on establishing plant species than bison. In some cases, mowing could have effects similar to grazing (Williams et al. 2007) if it preferentially decreases the abundance of dominant grasses. If managers initiate any of these regimes, they should simultaneously assess the abundance of invasive species during establishment (Isbell & Wilsey 2011; Wilsey et al. 2014). Martin & Wilsey (2006) did not find a greater number of exotic species seedlings than native species in this study, but this may not be found in every situation. Further work should be conducted with cattle as well as bison on prairie establishment and long-term persistence.

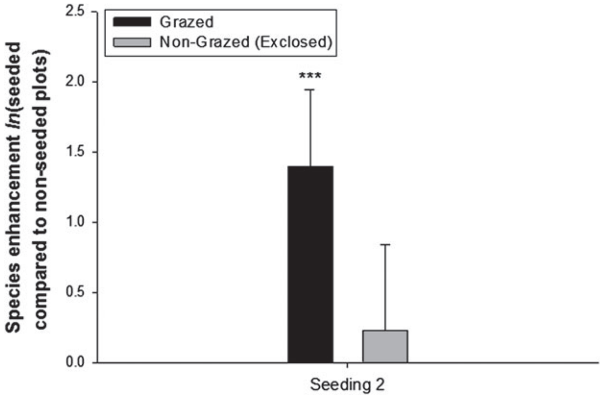

Figure 5. Rare species enhancement ln(seeded plots/control plots) from a 25-species seed addition in grazed and non-grazed plots (inside exclosures) in a tallgrass prairie restoration in central Iowa. Vertical bars are +1 standard error, and *** denotes significance from zero (i.e. null expectation of no enhancement, $p < 0.05$). The seed addition led to a 4-fold increase in abundance of added species ($e^{1.395} = 4.03$), but only when plots were grazed.

Acknowledgments

We would like to thank Pauline Drobney at Neal Smith National Wildlife Refuge for her support and encouragement, and Andrew Kaul for editorial help.

LITERATURE CITED

Allred BW, Fuhlendorf SD, Hamilton RG (2011a) The role of herbivores in Great Plains conservation: comparative ecology of bison and cattle. Ecosphere 2: art26, DOI: 10.1890/ES10-00152.1

Allred BW, Fuhlendorf SD, Engle DM, Elmore RD (2011*b*) Ungulate preference for burned patches reveals strength of fire-grazing interaction. Ecology and Evolution 1:132–144, DOI: 10.1002/ece3.12

Baer SG, Kitchen DJ, Blair JM, Rice CW (2002) Changes in ecosystem structure and function along a chronosequence of restored grasslands. Ecological Applications 12:1688–1701

Baer SG, Blair JM, Collins SL, Knapp AK (2004) Plant community responses to resource availability and heterogeneity during restoration. Oecologia 139:617–629

Bakker ES, Ritchie ME, Olff H, Milchunas DG, Knops JMH (2006) Herbivore impact on grassland plant diversity depends on habitat productivity and herbivore size. Ecology Letters 9:780–788

Bouressa EL, Doll JE, Cates RL, Jackson RD (2010) Burning and grazing to promote persistence of warm-season grasses sown into a cool-season pasture. Ecological Restoration 28:40–45

Bowles ML, Jones MD (2013) Repeated burning of eastern tallgrass prairie increases richness and diversity, stabilizing late successional vegetation. Ecological Applications 23:464–478

Camill P, McKone MJ, Sturges ST, Severud WJ, Ellis E, Limmer J, et al. (2004) Community- and ecosystem-level changes in species-rich tallgrass prairie restoration. Ecological Applications 14:1680–1694

Collins SL, Calabrese LB (2012) Effects of fire, grazing and topographic variation on vegetation structure in tallgrass prairie. Journal of Vegetation Science 23:563–575

Collins SL, Knapp AK, Briggs JM, Blair JM, Steinauer EM (1998) Modulation of diversity by grazing and mowing in native tallgrass prairie. Science 280:745–747

Copeland TF, Sluis W, Howe HF (2002) Fire season and dominance in an Illinois tallgrass prairie restoration. Restoration Ecology 10:315–323

Damschen EI, Haddad NM, Tewksbury JJ, Levey DJ (2006) Corridors increase plant species richness at large scales. Science 313:1284–1286

Dickson TL, Wilsey BJ, Busby RR, Gebhart DL (2009) *Melilotus officinalis* (yellow sweetclover) causes large changes in community and ecosystem processes in both the presence and absence of a cover crop. Biological Invasions 201:639–649

Eilers LJ, Roosa DM (1994) The vascular plants of Iowa. University of Iowa Press, Iowa City

Eriksson O, Ehrlen J (1992) Seed and microsite limitation of recruitment in plant populations. Oecologia 91:360–364

Foster BL (2001) Constraints on colonization and species richness along a grassland productivity gradient: the role of propagule availability. Ecology Letters 4:530–535

Foster BL, Dickson TL (2004) Grassland diversity and productivity: the interplay of resource availability and propagule pools. Ecology 85: 1541–1547

Foster BL, Gross KL (1997) Partitioning the effects of plant biomass and litter on *Andropogon gerardi* in old-field vegetation. Ecology 78:2091–2104

Foster BL, Dickson TL, Murphy CA, Karel IS, Smith VH (2004) Propagule pools mediate community assembly and diversity-ecosystem regulation along a grassland productivity gradient. Journal of Ecology 92:435–449

Fraser LH, Jentsch A, Sternberg M (2014) What drives plant species diversity? A global distributed test of the unimodal relationship between herbaceous species richness and plant biomass. Journal of Vegetation Science 25:1160–1166

Grime JP (1973) Competitive exclusion in herbaceous vegetation. Nature 242:344–347

Grman E, Bassett T, Brudvig LA (2013) Confronting contingency in restoration: management and site history determine outcomes of assembling prairies, but site characteristics and landscape context have little effect. Journal of Applied Ecology 50:1234–1243

Hartnett DC, Hickman KR, Walter LEF (1996) Effects of bison grazing, fire, and topography on floristic diversity in tallgrass prairie. Journal of Range Management 49:413–420

Henry M, Stevens H, Bunker DE, Schnitzer SA, Carson WP (2004) Establishment limitation reduces species recruitment and species richness as soil resources rise. Journal of Ecology 92:339–347

Howe HF (2000) Grass response to seasonal burns in experimental plantings. Journal of Range Management 53:437–441

Isbell FI, Wilsey BJ (2011) Increasing native, but not exotic, biodiversity enhances ecosystem functioning in ungrazed and intensely grazed grasslands. Oecologia 165:771–781

Jackson RD, Paine LK, Woodis JE (2010) Persistence of native C_4 grasses under high-intensity, short-duration summer bison grazing in the eastern tallgrass prairie. Restoration Ecology 18:65–73

Kagima B, Fairbanks WS (2013) Habitat selection and diet composition of reintroduced native ungulates in a fire-managed tallgrass prairie reconstruction. Ecological Restoration 31:79–88

Knapp AK, Blair JM, Briggs JM, Collins SL, Hartnett DC, Johnson LC, Towne EG (1999) The keystone role of bison in North American tallgrass prairie. BioScience 48:39–50

Knapp AK, Hoover DL, Blair JM, Buis G, Burkepile DE, Chamberlain A, et al. (2012) A test of two mechanisms proposed to optimize grassland aboveground primary production in response to grazing. Journal of Plant Ecology 5:357–365

Koerner SE, Collins SL (2014) Interactive effects of grazing, drought, and fire on grassland plant communities in North America and South Africa. Ecology 95:98–109

Kohl MT, Krausman PR, Kunkel K, DM Williams (2013) Bison versus cattle: are they ecologically synonymous? Rangeland Ecology and Management 6:721–731

Kucharik CJ, Fayram NJ, Cahill KN (2006) A paired study of prairie carbon stocks, fluxes, and phenology: comparing the world's oldest prairie restoration with an adjacent remnant. Global Change Biology 12:122–139

Lezama F, Baeza S, Altesor A, Cesa A, Chaneton EJ, Paruelo JM (2014) Variation of grazing-induced vegetation changes across a large-scale productivity gradient. Journal of Vegetation Science 25:8–21

Littell RC, Stroup WW, Freund RJ (2012) SAS for linear models. 2nd edition, Wiley and Sons, Cary, North Carolina

Martin LM, Wilsey BJ (2006) Assessing grassland restoration success: relative roles of seed additions and native ungulate activities. Journal of Applied Ecology 43:1098–1110

Martin LM, Wilsey BJ (2014) Native-species seed additions do not shift restored prairie plant communities from exotic to native states. Basic and Applied Ecology 15:297–304

Martin LM, Moloney KA, Wilsey BJ (2005) An assessment of grassland restoration success using species diversity components. Journal of Applied Ecology 42:327–336

McNaughton SJ (1979) Grazing as an optimization process: grass-ungulate relationships in the Serengeti. American Naturalist 113:691–703

McNaughton SJ, Milchunas DG, Frank DA (1996) How can net primary productivity be measured in grazing ecosystems? Ecology 77:974–977

Packard S, Mutel CF (1997) The tallgrass restoration handbook for prairies, savannas, and woodlands. Island Press, Washington, D.C.

Polley HW, Wilsey BJ, Derner JD (2005) Patterns of plant species diversity in remnant and restored tallgrass prairies. Restoration Ecology 13:480–487

Pywell RF, Bullock JM, Hopkins A, Walker KJ, Sparks TH, Burke MJW, Peel S (2002) Restoration of species-rich grassland on arable land: assessing the limiting processes using a multi-site experiment. Journal of Applied Ecology 39:294–309

Rabinowitz D, Rapp JK (1980) Seed rain in North American tall grass prairie. Journal of Applied Ecology 17:193–802

Sluis WJ (2002) Patterns of species richness and composition in re-created grassland. Restoration Ecology 10:677–684

Smith RS, Shiel RS, Millward D, Corkhill P (2000) The interactive effects of management on the productivity and plant community structure of an upland meadow: an 8-year field trial. Journal of Applied Ecology 37:1029–1043

Tilman D (1997) Community invasibility, recruitment limitation, and grassland biodiversity. Ecology 78:81–92

Towne EG, Hartnett DC, Cochran RC (2005) Vegetation trends in tallgrass prairie from bison and cattle grazing. Ecological Applications 15:1550–1559

Turnbull LA, Crawley MJ, Rees M (2000) Are plant populations seed-limited? A review of seed sowing experiments. Oikos 88:225–238

Williams DW, Jackson LL, Smith DD (2007) Effects of frequent mowing on survival and persistence of forbs seeded into a species-poor grassland. Restoration Ecology 15:24–33

Wilsey BJ (2010) Productivity and subordinate species response to dominant grass species and seed source during restoration. Restoration Ecology 18:628–637

Wilsey BJ, Polley HW (2003) Effects of seed additions and grazing history on diversity and productivity of subhumid grasslands. Ecology 84:920–931

Wilsey BJ, Parent G, Roulet NT, Moore TR, Potvin C (2002) Tropical pasture carbon cycling: relationships between C source/sink strength, above-ground biomass and grazing. Ecology Letters 5:367–376

Wilsey BJ, Daneshgar PP, Hofmockel K, Polley HW (2014) Invaded grassland communities have altered stability-maintenance mechanisms but equal stability compared to native communities. Ecology Letters 17:92–100

Xiong S, Nilsson C (1999) The effects of plant litter on vegetation: a meta-analysis. Journal of Ecology 87:984–994

Zobel M, Otsus M, Liira J, Moora M, Möls T (2000) Is small-scale species richness limited by seed availability or microsite availability? Ecology 81:3274–3282

Coordinating Editor: Brandon Bestelmeyer

Received: 11 July, 2014; First decision: 10 September, 2014; Revised: 29 January, 2015; Accepted: 30 January, 2015; First published online: 6 March, 2015